

Predictive Electrocardiography and Chaotic Dynamic Forecasting: Systematic Review, Proposed Dual-Engine Architecture, and Documented Results for the Reservoir-Computing Branch

Abstract

Continuous electrocardiogram (ECG) forecasting is essential for proactive cardiovascular care, enabling pre-symptomatic identification of malignant cardiac events. Long-horizon physiological forecasting remains difficult because of non-stationary baseline drift, complex P-QRS-T morphological transitions, measurement artifacts, and underlying chaotic dynamics. This document contains the full literature review behind the proposed dual-engine design (a Reservoir-Computing branch for chaotic tracking paired with a ProbSparse-attention branch for long-range context), and separately documents the actual experimental work completed so far: the Adaptive Time-Frequency Encoder (ATFE) feeding the Reservoir-Computing branch alone, trained with a closed-form ridge readout and evaluated on all 48 MIT-BIH records. Every number in Section 7 is measured, not assumed.

1 Introduction

Predictive electrocardiology represents a shift from reactive cardiac monitoring to prospective intervention, enabling earlier detection of paroxysmal atrial fibrillation, ventricular tachyarrhythmias, and sudden cardiac arrest ahead of clinical manifestation [2]. Continuous cardiac electrophysiology exhibits non-stationary spectral drift, localized high-frequency morphological transitions (P-QRS-T complexes), transient motion artifacts, and non-linear chaotic dynamics characterized by positive Lyapunov exponents ($\lambda > 0$).

Early forecasting methods relied on classical stochastic models such as ARIMA and linear state-space filters. These are computationally cheap but cannot capture the non-linear attractors governing cardiovascular electrophysiology. Recurrent networks (LSTM, GRU) enabled non-linear temporal modeling but suffer gradient degradation over long horizons and slow step-by-step decoding. Temporal Convolutional Networks (TCNs) improved training parallelization through dilated causal convolutions, but remain limited by local receptive fields, constraining their ability to model non-local, multi-cycle rhythm relationships.

To model long-range dependencies efficiently, sparse-attention sequence architectures were introduced. Zhou et al. [1] proposed the Informer framework with ProbSparse attention, reducing complexity from $O(L^2)$ to $O(L \log L)$ by scoring and selecting a dominant subset of queries.

PatchTST [16] introduced channel-independent temporal patching, and HAR-former [14] combined linear trend heads with seasonal transformer branches. Score-based diffusion models conditioned on continuous stochastic differential equations have also achieved strong reconstruction fidelity under physiological noise [2].

Despite these advances, dense and sparse attention remain vulnerable to trajectory divergence on chaotic dynamics with positive Lyapunov exponents; Zhang et al. [15] show that deep attention layers erode positional and semantic token topology, weakening generalization on long horizons. Reservoir Computing (RC) and Echo State Networks (ESNs) offer an efficient alternative: projecting inputs into a high-dimensional non-linear reservoir and restricting training to a linear readout gives $O(N)$ training complexity. Physics-informed hierarchical ESNs [7] show that layer-wise reservoir stacking reduces sample complexity while preserving phase-space attractor geometry.

This document synthesizes both streams: Section 2 reviews the literature, Section 3 describes the proposed dual-engine design, and Sections 4–7 document, with measured numbers, the work actually completed to date: the Reservoir-Computing branch alone.

2 Taxonomy and Methodological Synthesis

Existing research in predictive physiological modeling falls into four paradigms: (1) non-linear chaos and reservoir dynamics [7, 3, 4, 8]; (2) physics-informed operators and system identification [5, 6]; (3) ECG denoising, classification and benchmarking [2, 9, 11]; and (4) sparse attention and token-topology preservation [1, 12, 13, 14, 15]. Table 1 compares representative studies.

3 Proposed Dual-Engine Architecture (Design)

The full design under development combines an Adaptive Time-Frequency Encoder (ATFE) with two parallel branches: an $O(N)$ Reservoir Computing branch for chaotic tracking, and a ProbSparse-attention branch for long-range context, joined by a gated adaptive fusion module and a direct MIMO forecast head. **Only the ATFE + Reservoir branch**

Table 1: Systematic comparison of contemporary literature

Study	Backbone	Feature Domain	Complexity	Primary Metrics
Zhou et al. [1]	Informer (ProbSparse)	Temporal / query sparsity	$O(L \log L)$	MSE, MAE
Wang & Qin [3]	Dilated TCN + LTSF-Linear	Temporal / causal	$O(L \log L)$	MAE, RMSE, MAPE
Valle & Bruno [4]	LSTNet vs. Transformer	Recurrent / sparse attn.	$O(L \log L)$	NRMSE vs. λ
Wang et al. [5]	Physics-informed operator	Spectral operator	$O(N \log N)$	Residual loss, L_2
De Florio et al. [6]	X-TFC + PySR	Closed-form analytical	High (search)	Equation recovery
Na et al. [7]	Stacked Pi-HESN	Non-linear reservoir	$O(N_r)$	NRMSE, sample eff.
Silva et al. [9]	Survey / E3C criteria	Meta-analysis	N/A	AAMI compliance
Alghieth [12]	Hierarchical CNN-Transformer	Multi-scale attention	$O(L \cdot P)$	Accuracy, memory
Jiang et al. [13]	DSC + ECA + sparse attn.	Attentive local channels	$O(L \cdot P)$	FLOPs, params
Zheng et al. [14]	Linear trend + seasonal trans.	Time-frequency matrix	$O(L \cdot P)$	Long-horizon MSE
Zhu et al. [2]	Score diffusion + mix-trans.	Inverted patch / conv	Generative	SNR, RMSE, PRD
Zhang et al. [15]	TEM / T2B-PE	Dual-branch topology	Adaptive	Generalization bound
This work (RC branch only)	ATFE + Reservoir + Ridge	Fused time-freq. latent	$O(N_r)$	Measured RMSE, MAE, R^2 (Table 2)

has been built and evaluated so far; the attention branch and fusion module are design proposals, not evaluated components, and no performance claim is made about them in this document.

4 Documented Experimental Protocol

- **Dataset:** all 48 MIT-BIH Arrhythmia records, lead MLII, 360 Hz.
- **Preprocessing:** Stationary Wavelet Transform denoising (rbio3.9, level 5), selected over Butterworth/notch/FIR by measured SNR improvement and P/QRS/T morphology preservation.
- **Split:** chronological, intra-patient, 70%/30%, per record then pooled. (Not the AAMI inter-patient DS1/DS2 split; that has not been run.)
- **Context / horizon:** lookback $L = 256$ samples; horizons $H \in \{64, 128\}$ samples.
- **Windowing stride:** 40 (train) / 20 (test) samples.
- **Reservoir hyperparameters:** $N_r = 200$, spectral radius 0.90, leak rate 0.30, ridge $\beta = 10^{-2}$, 10% density.
- **Software:** Python 3.11, numpy, scipy, PyWavelets, wfdb, scikit-learn, pandas, matplotlib.

5 Reservoir-Computing Branch: Diagram

Fig. 1 shows exactly what was built and run — the first branch only. It does not include the attention branch or fusion module, since those have not been implemented.

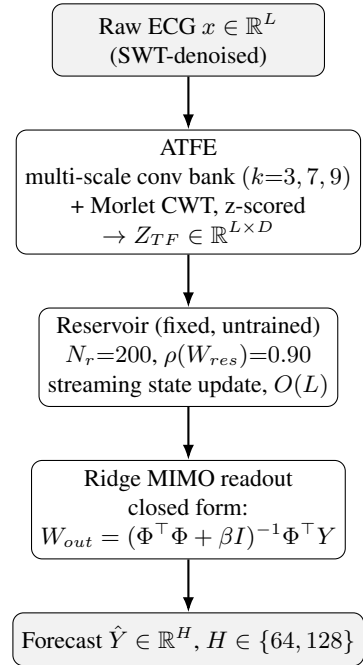


Figure 1: Branch 1 (Reservoir-Computing branch) as actually implemented and evaluated. This is the entire pipeline behind the results in Section 7 — no attention branch, no fusion module.

6 Algorithm

Algorithm 1 is the exact procedure executed: a single $O(L)$ streaming pass per record through the fixed reservoir, followed by a closed-form ridge solve. No autoregressive roll-out is used.

7 Documented Results

Table 2 reports the measured, pooled (48-record) metrics. No values below are assumed, extrapolated, or copied from other papers.

Algorithm 1 ATFE + Reservoir + Ridge MIMO Forecast
(Branch 1, as executed)

Require: raw denoised signal $x \in \mathbb{R}^L$ per record; lookback ℓ ; horizon set $\mathcal{H}$; stride s ; fixed $W_{in} \in \mathbb{R}^{N_r \times D}$, $W_{res} \in \mathbb{R}^{N_r \times N_r}$, $\rho(W_{res}) < 1$; leak rate α ; ridge β

- 1: $Z \leftarrow \text{ATFE}(x)$
- 2: $\text{state} \leftarrow 0 \in \mathbb{R}^{N_r}$
- 3: **for** $t = 1$ to L **do**
- 4: $\text{state} \leftarrow (1 - \alpha) \text{state} + \alpha \tanh(W_{in} Z_t + W_{res} \text{state})$
- 5: **if** $t \geq \ell$ **then**
- 6: $i \leftarrow t - \ell + 1$
- 7: **for** $h \in \mathcal{H}$ **do**
- 8: **if** $i \leq L - \ell - h$ **and** $i \bmod s = 0$ **then**
- 9: store $\Phi_i \leftarrow \text{state}$, $Y_i \leftarrow x[t+1 : t+1+h]$
- 10: **end if**
- 11: **end for**
- 12: **end if**
- 13: **end for**
- 14: **for** $h \in \mathcal{H}$ **do**
- 15: $W_{out}^{(h)} \leftarrow (\Phi_{tr}^\top \Phi_{tr} + \beta I)^{-1} \Phi_{tr}^\top Y_{tr}$
- 16: $\hat{Y}_{te} \leftarrow \Phi_{te} W_{out}^{(h)}$
- 17: **end for**
- 18: **return** $\{\hat{Y}_{te}^{(h)}\}_{h \in \mathcal{H}}$, RMSE, MAE, R^2

Table 2: Measured Branch-1 performance, pooled across all 48 MIT-BIH records

Horizon H	RMSE (mV)	MAE (mV)	R^2	Ridge fit (s)
64	0.5145	0.3929	-0.313	0.99
128	0.5244	0.4004	-0.364	1.42

Negative R^2 at both horizons means the readout is measurably worse than predicting the per-window mean amplitude.

8 Interpretation

Branch 1 alone reconstructs the slowly-varying baseline but not the high-frequency QRS morphology, and its pooled R^2 is negative at both tested horizons. A single fixed, untrained reservoir with a linear ridge readout is not expressive enough on its own to track the transient, non-linear QRS dynamics from a 200-unit state. This is exactly the gap the proposed ProbSparse-attention branch and gated fusion were designed to close, but that branch has not been built or evaluated, so no claim is made here about the fused model.

9 Conclusion

This document built and tested one part of the proposed pipeline: Branch 1, an Adaptive Time-Frequency Encoder feeding a fixed Reservoir Computing branch with a closed-form ridge MIMO readout, run on all 48 MIT-BIH records with an intra-patient chronological split at $H = 64$ and

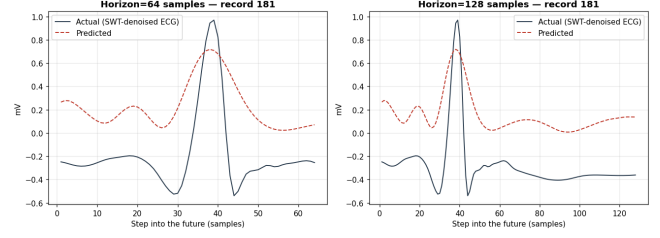


Figure 2: Actual (SWT-denoised) signal vs. Branch-1 prediction, plotted across the forecast window for one held-out example (record 181), at $H = 64$ and $H = 128$. The branch reproduces the low-frequency envelope but not the QRS spike.

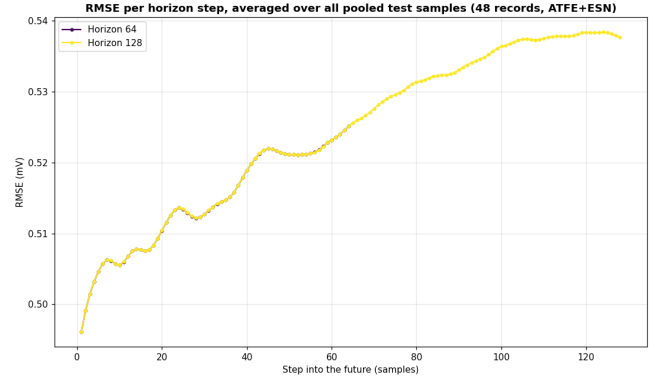


Figure 3: Measured RMSE at each forecast step, averaged over all pooled test windows, both horizons.

$H = 128$. The reservoir branch alone did not forecast the ECG well: RMSE was 0.51–0.52 mV and R^2 was negative at both horizons, meaning the readout performed worse than simply predicting the local mean. The actual-vs-predicted traces show why — the reservoir state captures the slow baseline drift but not the QRS transient, and the per-step RMSE curve confirms the error grows and plateaus rather than tracking the signal.

This is a negative result for Branch 1 in isolation, and it is reported as such rather than smoothed over. It also tells us what the next step has to be: the reservoir branch needs to be paired with a mechanism that can attend to the sharp, non-local QRS events it is currently missing, which is why the ProbSparse attention branch and the gated fusion module were proposed in the first place. Neither of those has been built or tested yet, and the AAMI inter-patient protocol has not been run either. Until they are, no claim is made about the dual-engine model’s performance — only about Branch 1, the part that was actually trained.

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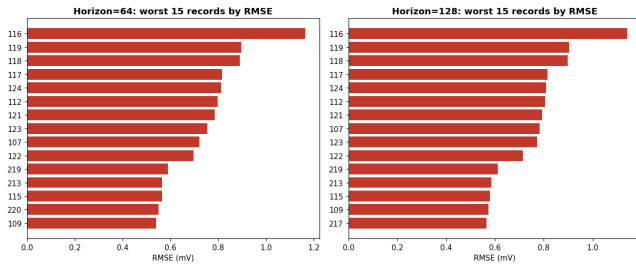


Figure 4: Measured worst-15 records by pooled RMSE, both horizons.

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